

EE596H (Optimization Techniques): Half Semester Examination
Total Questions: 3, Total Marks: 35, Time: 120 Mins

1. Consider the following problem: Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & \frac{1}{2}x_1^2 + \frac{1}{2}(x_2 - 3)^2 \\ \text{s.t.} \quad & x_1^2 - x_2 \leq 0, \\ & -x_1 + x_2 \leq 2. \end{aligned}$$

- (a) Is this a convex programming problem? Justify your answer. 2
- (b) Write down the KKT conditions and solve them to determine the KKT point. You will have to find Lagrange's (or KKT) multipliers at the solution as well. 7
- (c) Write the dual problem (No need to solve). Does strong duality hold? Explain and elaborate. 4

Total: 13

2. Consider the problem

$$\begin{aligned} \min \quad & \max\{x_1 + 2x_2, 3x_1 - x_2, -x_1 + x_2\} \\ \text{subject to} \quad & x_1 + x_2 = 1. \end{aligned}$$

- (a) Solve the above *min-max* problem manually or graphically. 7
- (b) Convert the problem to only *minimization problem* and solve it using KKT conditions. You will have to find Lagrange's (or KKT) multipliers at the solution as well. 8

Total: 15

3. Minimize $f(x_1, x_2) = 2x_1 - x_2 + x_1^2 + 3x_1x_2 + 3x_2^2$ using Steepest Descent algorithm. Take the initial point $x_1 = 0, x_2 = 1$. Run only the first two iterations. 7